

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F\,D$
T3•B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2}N_{r_1}{}^T C\,N_{r_1}M_F$	$-\text{New Scalar } L_{\alpha_1}\text{New Scalar } L_{\alpha_2}{}^2 M_F^2$ $+\frac{1}{2}\text{amb-dash Inv IN velt color } L_{\alpha_2}\text{New Scalar } L_{\alpha_1}{}^2\text{H scalar HGHG Inv } L_{\alpha_1-\alpha_2}\alpha_2\text{New velt color } L_{\alpha_1}{}^2\text{New velt color } L_{\alpha_2}{}^2$ $-\frac{1}{2}\text{amb-dash Inv IN velt color } L_{\alpha_2}\text{New Scalar } L_{\alpha_1}{}^2\text{H scalar HGHG Inv } L_{\alpha_1-\alpha_2}\alpha_2\text{New velt color } L_{\alpha_1}{}^2\text{New velt color } L_{\alpha_2}{}^2$	$-M_F^2\text{New velt color } L_{\alpha_1}\text{New velt color } L_{\alpha_2}{}^2$ $-\frac{1}{2}\text{amb-dash Inv IN velt color } L_{\alpha_2}\text{New Scalar } L_{\alpha_1}{}^2\text{H scalar HGHG Inv } L_{\alpha_1-\alpha_2}\alpha_2\text{New velt color } L_{\alpha_1}{}^2\text{New velt color } L_{\alpha_2}{}^2$	$+\frac{1}{2}iN_{r_1}{}^T C\,\gamma^{\mu_1}D_{\mu_1}(N_{r_1})$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 D$	$S_1 H$	$S_1 S_2$	$S_2 D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar1}_{a_1})$	$s(\text{lambda11Inv1}H_{a_2}\text{NewScalar1}_{a_1}\text{THiggsPortal})\text{CGInv1}_{a_1,a_2,a_2,a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$ $s(\text{lambda11Inv2}H_{a_2}\text{NewScalar1}_{a_1}\text{THiggsPortal})\text{CGInv2}_{a_1,a_2,a_2,a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$s(\text{lambda11Inv13NewScalar1}_{a_2}\text{NewScalar2}_{a_1}\text{TCrossScalarCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_1}\text{NewScalar2}_{a_1}^\dagger)$ $s(\text{lambda11Inv23NewScalar1}_{a_2}\text{NewScalar2}_{a_1}\text{TCrossScalarCGInv2}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_1}\text{NewScalar2}_{a_1}^\dagger)$	$+D_{\mu_1}(\text{NewScalar2}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar2}_{a_1})$

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} &+i\text{lambdaR2Im1}H_{a3}\text{NewScalar2}_{a4}\text{TBIGGPortalsCGInv1}_{a_1,a_2,a_4}H_{a_1}^\dagger\text{NewScalar2}_{a_2}^\dagger \\ &+i\text{lambdaR2Im2}H_{a3}\text{NewScalar2}_{a4}\text{TBIGGPortalsCGInv2}_{a_1,a_2,a_4}H_{a_1}^\dagger\text{NewScalar2}_{a_2}^\dagger \end{aligned}$	$\begin{aligned} &+\overline{\ell_{a_1,i_1}}P_RN_{r_1}\text{NewScalar1}_{a_1}y_{1,i_1,r_1} \\ &+\overline{y_{1,i_1,r_1}}N_{r_1}^TC_P\ell_{a_1,i_1}\text{NewScalar1}_{a_1}^\dagger \end{aligned}$	$\begin{aligned} &+\overline{\lambda_{T3}}\text{NewScalar2}_{a_1}\epsilon_{a_2,a_3}H_{a_1}^\dagger H_{a_2}^\dagger\text{NewScalar1}_{a_3}^\dagger \\ &+H_{a_1}H_{a_2}\text{NewScalar1}_{a_3}\lambda_{T3}\epsilon_{a_2,a_3}\text{NewScalar2}_{a_1}^\dagger \end{aligned}$	$\begin{aligned} &+\overline{y_{2,i_1,r_1}}N_{r_1}^TC_P\ell_{a_1,i_1}\text{NewScalar2}_{a_2}\bar{\epsilon}_{a_1,a_2} \\ &+\overline{\ell_{a_2,i_1}}P_RN_{r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{NewScalar2}_{a_1}^\dagger \end{aligned}$